# Lab: The Illusion Lab -- Why You See What You Expect

## Objective

Explore how **perception** works as prediction by testing optical illusions, sensory biases, and the role of prior expectations in shaping what you experience.

## Prerequisites

- Completed modules on Systems and Agents
- Read the module on Top-down Perception

## Part 1: Optical Illusion Gallery

**Goal**: Experience how your brain fills in predictions even when sensory data is ambiguous.

Look up or draw each of the following classic illusions. For each one, describe what you see and why your brain is "tricked."

| Illusion | What You See First | What Is Actually There | Why Your Brain Fills It In |
|---|---|---|---|
| Muller-Lyer (arrows on lines) | | | |
| Rubin's Vase (face/vase) | | | |
| The Dress (blue-black vs. white-gold) | | | |
| Kanizsa Triangle (invisible triangle) | | | |

Write your response here

## Part 2: The Prediction Test

**Goal**: Test how prior expectations change what you perceive.

Try this with a partner:

1. Partner A writes a sentence with a common phrase but replaces one word with a similar-sounding wrong word (e.g., "Paris in the the spring").
2. Partner B reads the sentence quickly.
3. Record whether Partner B noticed the error.

Repeat five times with different sentences. Track results:

| Sentence | Error Hidden | Partner Noticed? (Yes/No) | Why or Why Not? |
| --- | --- | --- | --- |
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

What does this tell you about the role of predictions in reading?

Write your response here

## Part 3: Sensory Deprivation Mini-Experiment

**Goal**: Observe what happens when a sensory channel is removed.

Safely try one of the following for 2 minutes, then write about your experience:

- **Option A**: Close your eyes and navigate a familiar room. How did your other senses compensate?
- **Option B**: Wear earplugs or noise-canceling headphones in a quiet setting. What did you notice that you normally miss?
- **Option C**: Eat a piece of food while holding your nose. How did the taste change?

Describe: 1. What prediction errors occurred when the sense was removed? 2. How did your brain try to compensate? 3. What does this reveal about how perception relies on multiple prediction streams?

*Write your response here*

## Part 4: Prediction Error Journal

**Goal**: Track moments of perceptual surprise over one day.

Record three moments where what you perceived did not match what you expected (a sound you misidentified, a person you mistook for someone else, a flavor that surprised you).

| Moment | What You Expected | What Actually Happened | How Your Brain Resolved It |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

Write your response here

## Summary Table

| **Concept** | **Definition** | **Your Example** |
|---|---|---|
| Prediction | The brain's best guess about incoming sensory data | |
| Prediction Error | The mismatch between prediction and actual sensory input | |
| Prior | Pre-existing belief or expectation that shapes perception | |
| Top-down Processing | Using internal models to interpret raw sensory signals | |